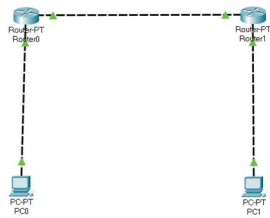


Practical No. 1

Study of Networking

Aim :- To study the IP address.

Circuit Diagram :-



Program :-

Router 0 > cli :-

```
enable
configure terminal
interface fastethernet 0/0
ip address 10.0.0.1 255.0.0.0
no shutdown
exit
interface fastethernet 0/1
ip address 20.0.0.1 255.0.0.0
no shutdown
exit
```

Router 1 > cli :-

```
enable
configure terminal
interface fastethernet 0/0
ip address 40.0.0.1 255.0.0.0
no shutdown
exit
interface fastethernet 0/1
ip address 20.0.0.2 255.0.0.0
no shutdown
exit
```

Pc 0 > Desktop > Ip configuration :-

```
ipv4 : 10.0.0.2
subnet mask : 255.0.0.0
Default gateway: 10.0.0.1
```

Pc 1 > Desktop > Ip configuration :-

ipv4 : 40.0.0.2

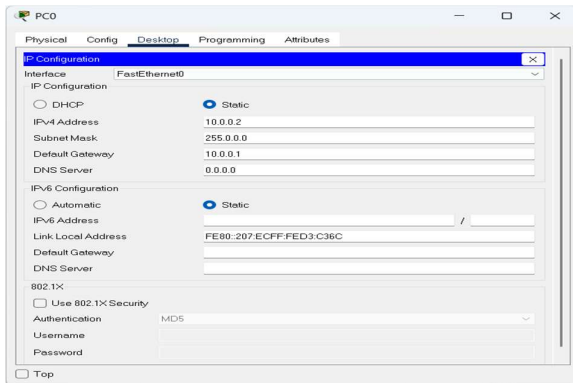
subnet mask : 255.0.0.0

Default gateway: 40.0.0.1

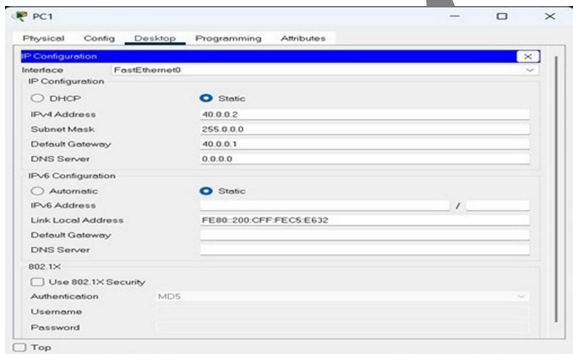
Program File Link :- <https://github.com/Dharni-dev1/Major-Dcn-practical-by-Pondharani-Devendra/blob/main/practical%20no.1.pkt>

Output :-

PC0 :-



PC1:-



Conclusion :- The program was executed successfully.

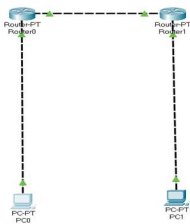
Practical No. 2

Study of Network Layer

a) Static Routing

Aim :- To study Static routing.

Circuit Diagram :-



Program :-

Router 0 > cli :-

```
enable
configure terminal
interface fastethernet 0/0
ip address 10.0.0.1 255.0.0.0
no shutdown
exit
interface fastethernet 0/1
ip address 20.0.0.1 255.0.0.0
no shutdown
exit
ip route 40.0.0.0 255.0.0.0 20.0.0.2
```

Router 1 > cli :-

```
enable
configure terminal
interface fastethernet 0/0
ip address 40.0.0.1 255.0.0.0
no shutdown
exit
interface fastethernet 0/1
ip address 20.0.0.2 255.0.0.0
no shutdown
exit
ip route 10.0.0.0 255.0.0.0 20.0.0.1
```

Pc 0 > Desktop > Ip configuration :-

ipv4 : 10.0.0.2

subnet mask : 255.0.0.0

Default gateway: 10.0.0.1

Pc 1 > Desktop > Ip configuration :-

ipv4 : 40.0.0.2

subnet mask : 255.0.0.0

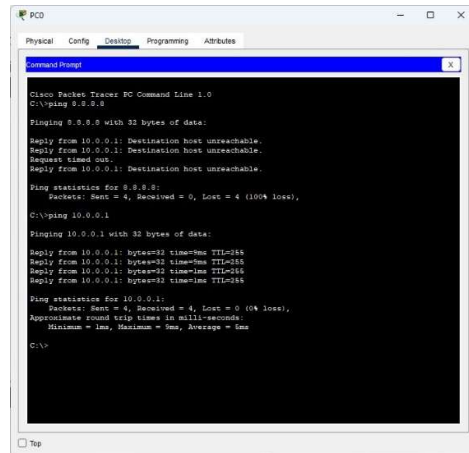
Default gateway: 40.0.0.1

Program File Link :-

<https://github.com/Dharni-dev1/Major-Dcn-practical-by-Pondharani-Devendra/blob/main/practical%20no.2.pkt>

Output :-

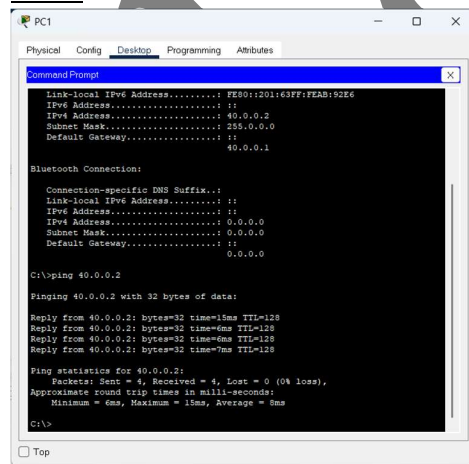
PC0 :-



```

PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 10.0.0.1: Destination host unreachable.
Request timed out.
Reply from 10.0.0.1: Destination host unreachable.
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 10.0.0.1
Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time=6ms TTL=64
Reply from 10.0.0.1: bytes=32 time=6ms TTL=64
Reply from 10.0.0.1: bytes=32 time=6ms TTL=64
Reply from 10.0.0.1: bytes=32 time=6ms TTL=64
Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 6ms, Average = 6ms
C:\>
  
```

PC1 :-



```

PC1
Physical Config Desktop Programming Attributes
Command Prompt
Link-local IPv6 Address . . . . . FE80::201:63FF:FEAB:92E6
IPv6 Address . . . . . ::
IPv6 Address . . . . . 40.0.0.2
Subnet Mask . . . . . 255.0.0.0
Default Gateway . . . . . 40.0.0.1

Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . . ::
IPv6 Address . . . . . ::
Subnet Mask . . . . . 0.0.0.0
Default Gateway . . . . . 0.0.0.0

C:\>ping 40.0.0.2
Pinging 40.0.0.2 with 32 bytes of data:
Reply from 40.0.0.2: bytes=32 time=15ms TTL=128
Reply from 40.0.0.2: bytes=32 time=6ms TTL=128
Reply from 40.0.0.2: bytes=32 time=6ms TTL=128
Reply from 40.0.0.2: bytes=32 time=7ms TTL=128
Ping statistics for 40.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 15ms, Average = 6ms
C:\>
  
```

Conclusion :- The program was executed successfully.

b) RIP Routing

Aim :- To study RIP routing.

Circuit Diagram :-



Program :-

Router 0 > cli :-

```
enable
configure terminal
interface fastethernet 0/0
ip address 10.0.0.1 255.0.0.0
no shutdown
exit
interface fastethernet 0/1
ip address 20.0.0.1 255.0.0.0
no shutdown
exit
router rip
version 2
network 10.0.0.0
network 20.0.0.0
exit
```

Router 1 > cli :-

```
enable
configure terminal
interface fastethernet 0/0
ip address 40.0.0.1 255.0.0.0
no shutdown
exit
interface fastethernet 0/1
ip address 20.0.0.2 255.0.0.0
no shutdown
exit
router rip
```

```

version 2
network 40.0.0.0
network 20.0.0.0
exit

```

Pc 0 > Desktop > Ip configuration :-

```

ipv4 : 10.0.0.2
subnet mask : 255.0.0.0
Default gateway: 10.0.0.1

```

Pc 1 > Desktop > Ip configuration :-

```

ipv4 : 40.0.0.2
subnet mask : 255.0.0.0
Default gateway: 40.0.0.1

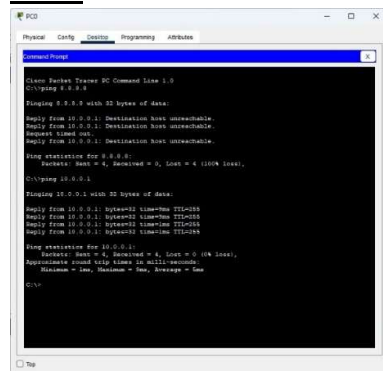
```

Program File Link :-

<https://github.com/Dharni-dev1/Major-Dcn-practical-by-Pondharani-Devendra/blob/main/practical%20no.2b.pkt>

Output :-

PC0:-



```

C:\Users\PC0>ping 10.0.0.1
Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Request timed out.
Reply from 10.0.0.1: Destination host unreachable.

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
    Approximate round trip times in milliseconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

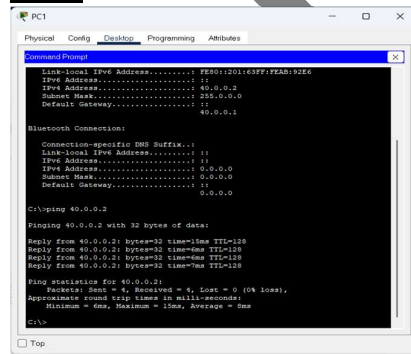
C:\Users\PC0>ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\PC0>

```

PC1:-



```

C:\Users\PC1>ipconfig
Link-local IPv6 Address . . . . . FE80::D01:65FF:FEA0:9266
IPv6 Address. . . . . ::
IPv6 Subnet Mask . . . . . ::
Default Gateway . . . . . ::

Bluetooth Connection:
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . ::
IPv6 Address. . . . . ::
IPv6 Subnet Mask . . . . . ::
Default Gateway . . . . . ::

C:\Users\PC1>ping 40.0.0.2
Pinging 40.0.0.2 with 32 bytes of data:
Reply from 40.0.0.2: bytes=32 time=1ms TTL=128
Reply from 40.0.0.2: bytes=32 time=1ms TTL=128
Reply from 40.0.0.2: bytes=32 time=1ms TTL=128
Reply from 40.0.0.2: bytes=32 time=1ms TTL=128

Ping statistics for 40.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\PC1>

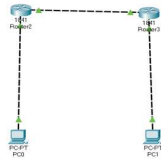
```

Conclusion :- The program was executed successfully.

c) OSPF Routing

Aim :- To study OSPF routing.

Circuit Diagram :-



Program :-

Router 0 > cli :-

```
enable
configure terminal
interface fastethernet 0/0
ip address 10.0.0.1 255.0.0.0
no shutdown
exit
interface fastethernet 0/1
ip address 20.0.0.1 255.0.0.0
no shutdown
exit
router ospf 1
network 10.0.0.0 0.255.255.255 area 0
network 20.0.0.0 0.255.255.255 area 0
exit
```

Router 1 > cli :-

```
enable
configure terminal
interface fastethernet 0/0
ip address 40.0.0.1 255.0.0.0
no shutdown
exit
interface fastethernet 0/1
ip address 20.0.0.2 255.0.0.0
no shutdown
exit
router ospf 2
network 40.0.0.0 0.255.255.255 area 0
```

network 20.0.0.0 0.255.255.255 area 0
exit

Pc 0 > Desktop > Ip configuration :-

ipv4 : 10.0.0.2

subnet mask : 255.0.0.0

Default gateway: 10.0.0.1

Pc 1 > Desktop > Ip configuration :-

ipv4 : 40.0.0.2

subnet mask : 255.0.0.0

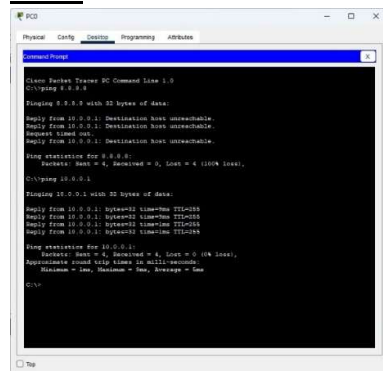
Default gateway: 40.0.0.1

Program File Link :-

<https://github.com/Dharni-dev1/Major-Dcn-practical-by-Pondharani-Devendra/blob/main/practical%20no.2c.pkt>

Output :-

PC0:-



```

C:\Users\PC0>ipconfig /all

Ethernet adapter Ethernet0:

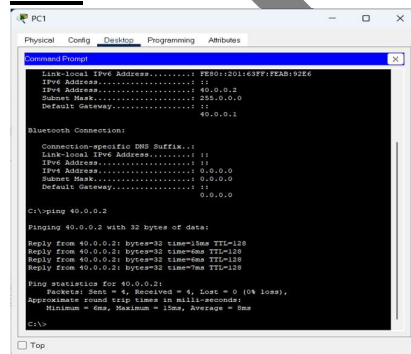
   Connection-specific DNS Suffix . : 
   . . . . .
   IPv4 Address. . . . . : 10.0.0.2
   Subnet Mask . . . . . : 255.0.0.0
   Default Gateway . . . . . : 10.0.0.1

C:\Users\PC0>ping 40.0.0.2

Pinging 40.0.0.2 with 32 bytes of data:
Reply from 40.0.0.2: Destination host unreachable.
Reply from 40.0.0.2: Destination host unreachable.
Reply from 40.0.0.2: Destination host unreachable.
Reply from 40.0.0.2: Destination host unreachable.

Ping statistics for 40.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
  
```

PC1:-



```

C:\Users\PC1>ipconfig /all

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix . : 
   . . . . .
   IPv4 Address. . . . . : 40.0.0.2
   Subnet Mask . . . . . : 255.0.0.0
   Default Gateway . . . . . : 40.0.0.1

C:\Users\PC1>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
  
```

Conclusion :- The program was executed successfully.

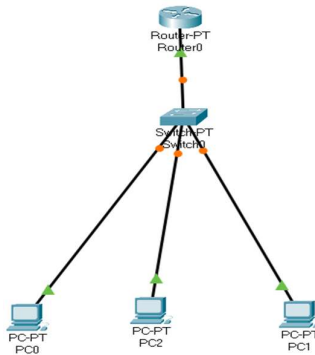
Practical No. 3

Study of Application Layer

a) DHCP

Aim :- To study DHCP.

Circuit Diagram :-



Program :-

Router 0 > cli :-

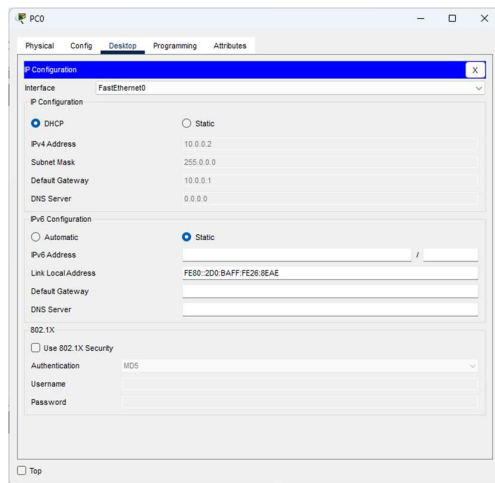
```
enable
configure terminal
interface fastethernet 0/0
ip address 10.0.0.1 255.0.0.0
no shutdown
exit
ip dhcp pool syit
network 10.0.0.0 255.0.0.0
default-router 10.0.0.1
ip dhcp excluded-address 10.0.0.2 10.0.0.1
ip dhcp pool syit
network 10.0.0.0 255.0.0.0
default-router 10.0.0.1
ip dhcp excluded-address 10.0.0.2 10.0.0.13
exit
```

Program File Link :-

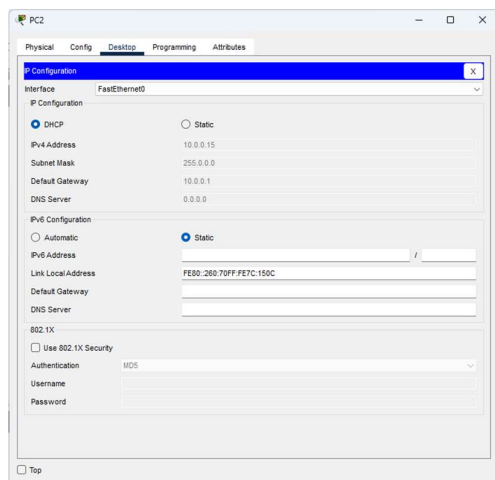
<https://github.com/Dharni-dev1/Major-Dcn-practical-by-Pondharani-Devendra/blob/main/practical%20no.3a.pkt>

Output:-

Pc0



Pc1

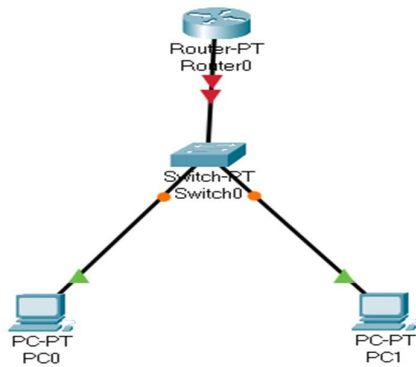


Conclusion :- The program was executed successfully.

b) DNS

Aim :- To study DNS.

Circuit Diagram :-



Program :-

Router 0 > cli :-

```
enable
configure terminal
interface fastethernet 0/0
ip address 10.0.0.1 255.0.0.0
no shutdown
exit
ip dhcp pool syit
network 10.0.0.0 255.0.0.0
default-router 10.0.0.1
dns-server 8.8.8.8
exit
ip host www.syit.com 10.0.0.1
exit
```

Pc 0 > Desktop > Ip configuration :-

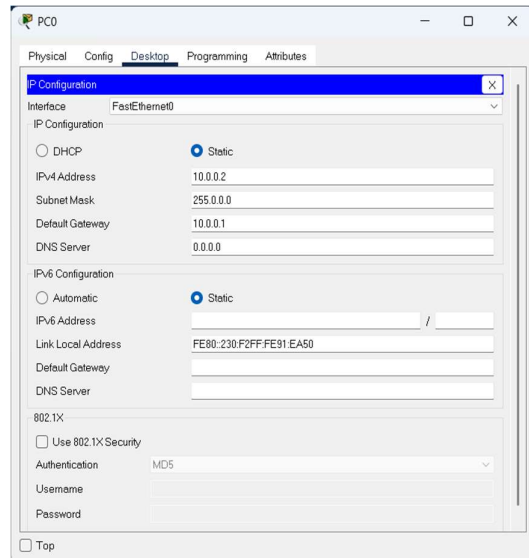
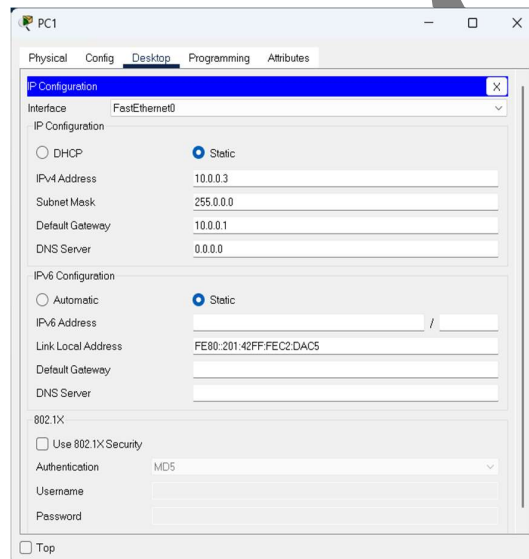
```
ipv4 : 10.0.0.2
subnet mask : 255.0.0.0
Default gateway: 10.0.0.1
```

Pc 1 > Desktop > Ip configuration :-

```
ipv4 : 10.0.0.3
subnet mask : 255.0.0.0
Default gateway: 10.0.0.1
```

Program File Link :-

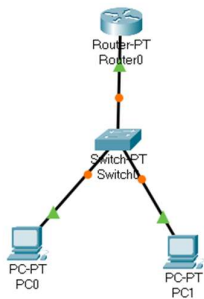
<https://github.com/Dharni-dev1/Major-Dcn-practical-by-Pondharani-Devendra/blob/main/practical%20no.3b.pkt>

Output:-**PC0****PC1**

Conclusion:-The program executed Successfully.

c) FTP

Aim :- To study FTP.



Program :-

Router 0 > cli :-

```

enable
configure terminal
interface fastethernet 0/0
ip address 10.0.0.1 255.0.0.0
no shutdown
exit
ip ftp username bscit
ip ftp password syit
ip ftp server enable
exit
  
```

Pc 0 > Desktop > Ip configuration :-

```

ipv4 : 10.0.0.2
subnet mask : 255.0.0.0
Default gateway: 10.0.0.1
  
```

Pc 1 > Desktop > Ip configuration :-

```

ipv4 : 10.0.0.3
subnet mask : 255.0.0.0
Default gateway: 10.0.0.1
  
```

Pc 0 > Desktop > Text Editor :-

This is from PC0

Pc 0 > Desktop > Command Prompt :-

```

ftp 10.0.0.1
username : bscit
password : syit
put testfile.txt
  
```

Pc 1 > Desktop > Command Prompt :-

ftp 10.0.0.1

username : bscit

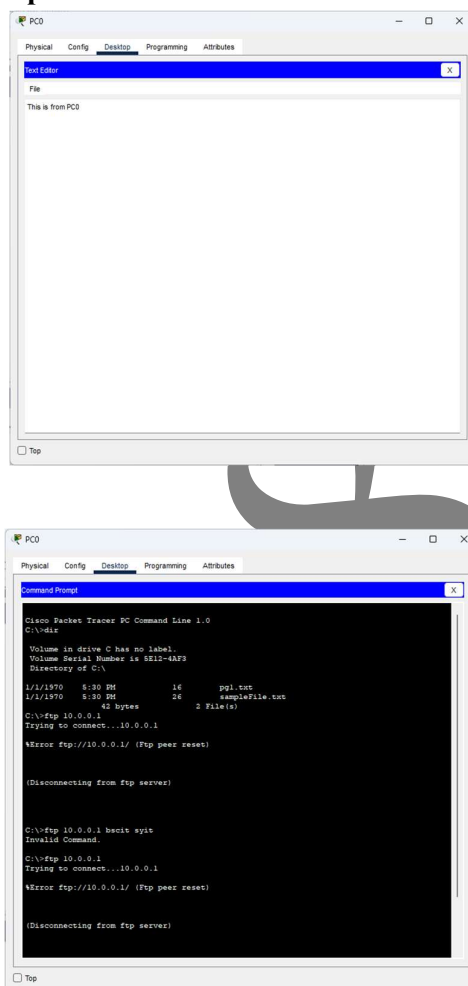
password : syit

get testfile.txt

Program File Link :-

<https://github.com/Dharni-dev1/Major-Dcn-practical-by-Pondharani-Devendra/blob/main/practical%20no.3c.pkt>

Output:-



Conclusion:-The program executed Successfully.

d) HTTP

Aim :- To study HTTP.

Circuit Diagram:-



Program :-

Server 0 > Desktop > Ip configuration :-

ipv4 : 10.0.0.2

subnet mask : 255.0.0.0

Default gateway: 0.0.0.0

Server 0 > Services > HTTP :-

HTTP : on

HTTPS : on

Server 0 > Services > TFTP :-

service: off

Server 0 > Services > HTTP > index.html > Edit :-

```
<html>
```

```
<head>
```

```
<title>SYIT</title>
```

```
</head>
```

```
<body>
```

```
<h1>I am SYIT</h1>
```

```
</body>
```

```
</html>
```

Pc 0 > Desktop > Ip Configuration :-

ipv4 : 10.0.0.3

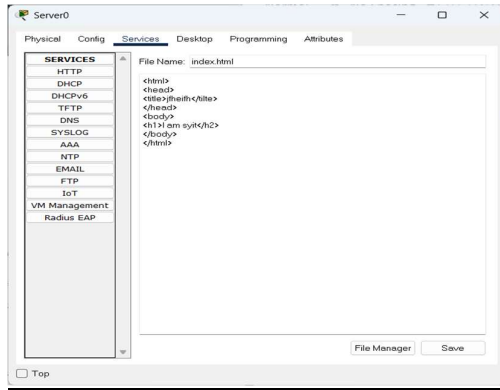
subnet mask : 255.0.0.0

Default gateway: 10.0.0.2

Program File Link :-

<https://github.com/Dharni-dev1/Major-Dcn-practical-by-Pondharani-Devendra/blob/main/practical%20no.3d.pkt>

Output:-

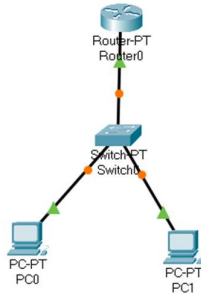


Conclusion:-The program executed Successfully.

e) TELNET

Aim :- To study TELNET.

Circuit diagram:-



Program :-

Router 0 > cli :-

```

enable
configure terminal
interface fastethernet 0/0
ip address 10.0.0.1 255.0.0.0
no shutdown
exit
line vty 0 10
password syit
login
transport input telnet
exit
exit
  
```

Pc 0 > Desktop > Ip configuration :-

```

ipv4 : 10.0.0.2
subnet mask : 255.0.0.0
Default gateway: 10.0.0.1
  
```

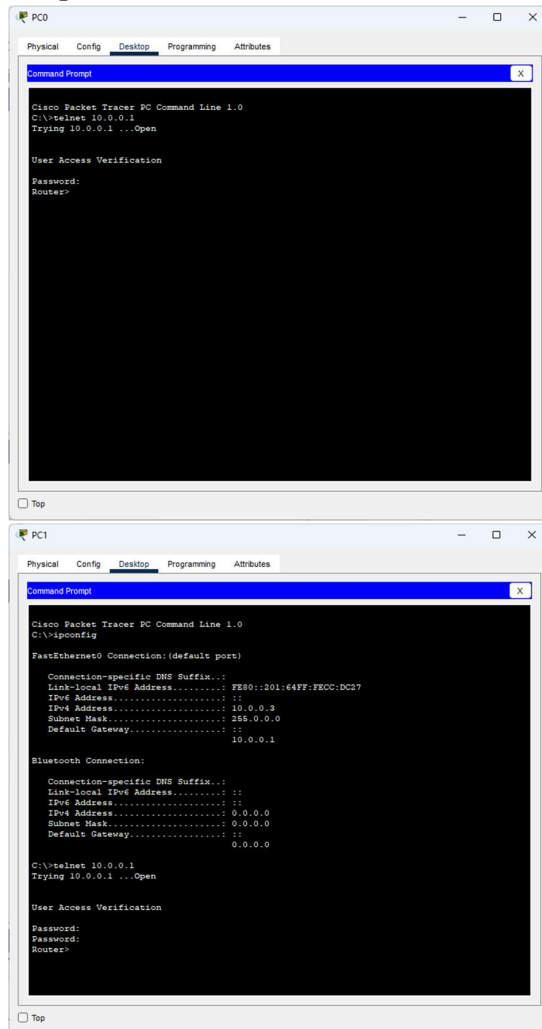
Pc 1 > Desktop > Ip configuration :-

```

ipv4 : 10.0.0.3
subnet mask : 255.0.0.0
Default gateway: 10.0.0.1
  
```

Program File Link :-

<https://github.com/Dharni-dev1/Major-Dcn-practical-by-Pondharani-Devendra/blob/main/practical%20no.3e.pkt>

Output:-

```
PC0
-----
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 10.0.0.1
Trying 10.0.0.1 ...Open

User Access Verification
Password:
Router>

PC1
-----
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)
Connection-specific DNS Suffix...:
Link-Local IPv6 Address...: FE80::201:64FF:FECC:DC27
IPv4 Address...:
IPv4 Address...: 10.0.0.3
Subnet Mask...: 255.0.0.0
Default Gateway...:
10.0.0.1

Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-Local IPv6 Address...:
IPv4 Address...:
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...:
0.0.0.0

C:\>telnet 10.0.0.1
Trying 10.0.0.1 ...Open

User Access Verification
Password:
Router>
```

Conclusion:-The program executed Successfully.